Osafu Augustine Egbon, Ph.D.

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Education

MBA in Data Science and Analytics
University of São Paulo, São Carlos, Brazil — *January 2024*
Capstone Project: *Neural Network Architecture for Mapping the Spatial and Temporal Patterns of Malaria Among Under-Five Children in Nigeria*.

Ph.D. in Statistics (Joint degree)
University of São Paulo, São Carlos & Federal University of São Carlos, São Carlos, Brazil — *July 2023*
Dissertation: *Bayesian Spatial Process Models for Activation Patterns in Transcranial Magnetic Stimulation Mapping*.

M.Tech in Statistics
The Federal University of Technology, Akure, Nigeria — *October 2018*
Thesis: *Application of Lee-Carter Model for Fertility Projection in Africa*.

B.Tech in Statistics
The Federal University of Technology, Akure, Nigeria — *April 2014*
Project: *Application of Discriminant Analysis in Modeling Students' Academic Performance*.

Awards/Fellowships

- **NIH Award to Mentor Summer Fellows**, National Institutes of Health, USA — *February 2025*
  NIH competitive award to mentor summer fellows in the field of Biostatistics and AI for studying Cancer developmental trajectories at single-cell resolution.
- **Honorable Mention – Outstanding Ph.D. Dissertation Award**, University of São Paulo, Sao Paulo, Brazil — *November 2024*
  Competitive award to recognize excellence in doctoral research in the field of Exact and Earth Sciences.
- **Visiting Scholar Fellowship**, Ludwig-Maximilians-Universität München, Germany, — *October 2022 – June 2023*
  Sponsored by **CAPES (Coordination for the Improvement of Higher Education Personnel), Brazil** to support collaborative international research in Brain mapping for depression treatment.
- **Ph.D. Scholarship**, University of Sao Carlos and University of São Paulo, Brazil — *July 2019 – September 2022*
  Awarded by **CAPES, Brazil** for doctoral studies and interdisciplinary research in spatial transcriptomics and health disparities.
- **Outstanding Academic Performance Award**, The Federal University of Technology Akure, Nigeria — *April 2014*
  Presented by the **Association of Statistics Students of Nigeria** for graduating with top honors in Statistics.

Grants and Funding

Modeler, Vaccine Impact Modeling Consortium (VIMC) Grant — *2023–2024*
Project title: Model-based geospatial estimation of vaccine impact and child health disproportionality in low and middle-income countries (LMICs).
Total award: €120,000

Research Experience

Postdoctoral researcher — *October 2023 — present*
National Institute of Environmental Health Sciences, NIEHS, NIH, Durham, USA
- Develop visualization and statistical methods for studying the impact of therapeutic interventions in melanoma cancer progression at single-cell resolution.
- Develop AI tools for integrating and analyzing multiple spatial omics data for studying the developmental process of colitis.
- Publish scientific research papers and present biological findings at national and international conferences, and poster presentations at NIEHS Scientific events.

- Led to publications in *Briefings in Bioinformatics*, a highly reputable journal in Computational Biology.

Modeler (Funded by the Vaccine Impact Modeling Consortium, London, United Kingdom) *September 2023 – August 2024*
The Federal University of Technology Akure, Akure, Nigeria.

- Developed statistical and mathematical models to estimate vaccine impact among children in Low and Middle-income countries (LMICs).
- Produced estimates of geospatial health patterns before and after public health interventions and quantified the impact of mothers' and caregivers' decision-making status on child vaccination.
- Collaborated with global partners and presented findings identifying regions with low vaccine coverage and the associated risk factors.
- Mentored 3 team members on modeling and computational tools for analyzing large Demographic and Health Survey vaccination datasets of developing countries
- Publish findings in scholarly articles in reputable journals.

Modeler *September 2022- April 2024*
Institute of Mathematics and Computer Sciences, University of São Paulo, São Paulo, Brazil

- Led an interdisciplinary research team, bringing together collaborators from diverse scientific backgrounds. Directed efforts to analyze and map spatial patterns of critical public health issues, with a focus on the effects of conflict on the health outcomes of children and women in developing countries.
- Developed spatial statistical models to provide mapping estimates of the impact of fatalities due to armed conflicts in West and Central Africa, and the impact on public health interventions.
- Led to the publication of scholarly articles in *Spatial Statistics* and *Scientific Reports* journals.

Research Associate – Survival and Reliability Research Group *October 2021 – July 2023*
Institute of Mathematics and Computer Sciences, University of São Paulo, São Paulo, Brazil

- Designed novel statistical models for predicting mechanical part lifespan in Brazilian railways.
- Developed Shiny application interfaces for user-friendly implementation of reliability models.
- Published scholarly articles in reputable journals and presented at international scientific conferences.
- Co-led the research group of 5 members to achieve over 4.50% improvement in predictive accuracy of parts failure compared with traditional methods.
- Led to a publication in the esteemed *Brazilian Journal of Probability and Statistics*.

Visiting Research scholar *October 2022 — June 2023*
Institute of Statistics, Ludwig Maximilian University of Munich, Munich, Germany

- Developed novel non-parametric joint spatial models to borrow and share information across multiple subjects in order to separate shared and subject-specific response effects to TMS.
- Substantially contributed scientific knowledge and experience in a group tasked with developing survival models.
- Led to a publication in the *Journal of the Royal Statistical Society, Series C*, a highly reputable journal in the field of Statistics.

Research Associate – Biostatistics and Spatial Modeling Group *January 2019 – July 2023*
Department of Statistics, Federal University of Technology, Akure, Nigeria

- Developed Bayesian geospatial statistical models to identify risk factors of malnutrition, anemia, malaria, and similar ailments among women and children in developing countries, and map the spatial patterns to address public health concerns.
- Present analytical and quantitative findings to the committee of public health policy stakeholders in Nigeria.
- Published scholarly articles in reputable public health journals that attracted *significant media attention in Nigeria*, highlighting the societal relevance and policy implications of the research findings (link).

Teaching Experience

Instructor – Introduction to Statistical Inference *October 2024 & March 2025*
Biostatistics and Computational Biology Branch, National Institute of Environmental Health Sciences (NIEHS), NIH, Durham, NC, USA

- Prepared course content and delivered weekly sessions on the *Statistical Inference* course to a total of approximately 16 NIEHS fellows and junior biomedical researchers per class.
- Provided instruction on hypothesis testing, confidence intervals, regression analysis, and prediction, and their real-world biomedical applications.
- Designed exercises and datasets on R-markdown for hands-on understanding and application of statistical concepts.

Adjunct Lecturer - Probability Theory and Spatial Statistics courses *September 2024 – present*
Department of Statistics, The Federal University of Technology Akure, Akure, Nigeria

- Develop, teach, and assess undergraduate and graduate courses: *Probability Theory* and *Spatial Statistical Inference* courses
- Develop course materials for introductory courses in statistics.

- Attract research grant through collaborative development of research proposals.

Teaching Assistant – Probability Theory Courses *June 2021 – June 2022*
Institute of Mathematics and Computer Sciences, University of São Paulo, Brazil
- Supported teaching and assessment of approximately 30 students in two undergraduate courses: *Introduction to Probability Theory* & *Probability Theory II*
- Conducted tutorials, helped reinforce concepts, and jointly brainstormed with students in solving challenging exercises with practical applications.
- Conducted one-on-one meetings to address students' concerns.

High School Mathematics Educator *February 2017 – April 2019*
Oredo Girls Senior Secondary School, Benin City, Nigeria
- Delivered engaging mathematics instruction to 4 high school classes, averaging 45 students per class, over a full academic session.
- Developed and implemented structured lesson plans, evaluated student performance, and provided individualized academic support to enhance learning outcomes.
- Mentored students in preparation for national standardized examinations and competitive mathematics contests, fostering critical thinking and problem-solving skills.

Teaching Assistant - Statistics for Bioscience course *January 2018 — December 2018*
Department of Statistics, The Federal University of Technology, Akure, Nigeria
- Delivered lectures and tutorial sessions for the course *Introduction to Statistics for Biological Sciences I & II*, designed for first-year undergraduate students from statistics and second-year undergraduate students from life science departments.
- Instructed an average of 65 students per semester, ensuring foundational understanding of key statistical concepts such as descriptive statistics, probability distributions, hypothesis testing, and linear regression.
- Assisted in developing course materials, including lecture notes, problem sets, and practical exercises using R and Microsoft Excel.
- Provided one-on-one academic support, conducted review sessions before exams, and supported student learning through office hours.
- Collaborated with faculty members to grade assignments and exams, and offered feedback to students to enhance their academic performance.

High School Mathematics Educator *October 2015 — January 2017*
Word of Faith Schools, Benin City, Nigeria.
- Delivered engaging and interactive mathematics instruction to a total of 72 high school students per academic session, fostering critical thinking and problem-solving skills.
- Designed and implemented effective lesson plans aligned with the national curriculum, integrating real-world applications to enhance student understanding.
- Assessed student performance through classwork, homework, and examinations, and provided constructive feedback to support academic growth.
- Mentored students for academic competitions; notably, one student advanced through local and state rounds to place 3rd at the national level.

Mentorship Experience

National Institute of Environmental Health Sciences, NIEHS, NIH, Durham, NC, USA *September 2024 — present*
Mentor, Master's Program in Mathematics, University of Ghana, Accra, Ghana
- Serving as the official academic advisor for a Master's student developing a deep learning framework to predict malnutrition prevalence among children in Ghana using national health survey and environmental data.

Co-mentor, Ph.D. Program in Statistics, Federal University of Technology, Akure, Nigeria
- Supporting a second-year doctoral student in designing a novel spatial statistical model to map malaria incidence and vaccination uptake across Nigeria, accounting for prevalence shocks due to armed conflict and population displacement.

Co-mentor, Ph.D. Program in Statistics, Federal University of Technology, Akure, Nigeria
- Guiding a first-year doctoral student in developing computational methods to identify biomarkers and molecular pathways associated with Plasmodium infections, using spatial transcriptomics and single-cell RNA sequencing data.

Co-mentor, Ph.D. Program in Statistics, University of São Paulo, São Carlos, Brazil
- Guiding a second-year doctoral student in developing spatial models for mapping Avian Influenza in Chile.

Institute of Mathematics and Computer Sciences, University of Sao Paulo, Sao Paulo, Brazil *September 2023 — June 2024*
Mentor, Master's Program in Mathematics, University of Ghana, Accra, Ghana
- Served as the official academic advisor for a Master's student who developed machine learning algorithms to predict malaria prevalence in Ghana. The student is now serving as a data science analyst in a reputable research institute in Ghana.

Co-mentor, Master's Program in Statistics, Federal University of Technology, Akure, Nigeria
- Guided a Master's student who developed a spatial statistical model to predict a geospatial map of undernutrition among children in Nigeria to identify high-burden population clusters for policy responses. The student is now a spatial modeler in an epidemiological research institute in Nigeria.

Leadership Position

Co-Chair, *NIEHS Biomedical Career Symposium, National Institute of Environmental Health Sciences (NIEHS) September 2025 – April 2026*

- Provided strategic leadership for the planning and execution of the 2026 NIEHS Biomedical Career Symposium.
- Designed and disseminated trainee surveys to assess needs and guide program development.
- Coordinated the invitation and engagement of keynote speakers, workshop facilitators, and career panelists from academia, industry, government, and nonprofit sectors.
- Recruited and managed volunteers to ensure seamless execution of event logistics.

Session Chair, *Advances in Generative Modeling session*, Joint Statistical Meeting (JSM), American Statistical Association, Nashville, Tennessee, USA. *August 2025*

- Presented novel spatiotemporal statistical methods for single-cell data to analyze in multiple perturbation scenarios.
- Facilitated scholarly discussion and led a session focused on cutting-edge developments in generative modeling applications in biomedical research.

Co-Chair, *NIEHS Trainee Assembly (NTA), National Institute of Environmental Health Sciences (NIEHS)*, *January 2024 – February 2025*

- Represent and advocate for over 160 postdoctoral and predoctoral fellows in an official capacity at the National Institute of Environmental Health Sciences (NIEHS).
- Organize major career development events.
- Present fellow's concern to NIEHS management and work closely with the NIH Office of Intramural Training & Education (OITE).
- Lead and coordinate monthly NTA Steering Committee meetings, fostering collaboration and enhancing professional development opportunities for fellows.

Travel Awards

- **Travel Award – Advanced Imaging Workshop**, Boston USA *May 2024*
  Sponsored by the **American Society for Cell Biology (ASCB)** to attend an international imaging workshop focused on cutting-edge cell biology techniques.
- **Travel Award – Vaccine impact modeling workshop**, London, UK, *September 2023*
  Sponsored by the **Vaccine Impact Modeling Consortium**, London, UK.

Developed shiny dashboard and package

MalariaApp, 2025 (link) For malaria incidence and prevalence in Nigeria: it estimates uncertainties of malaria incidence and prevalence
TMSBrainApp, 2023 (link) For TMS brain mapping. It adopts Bayesian spatial statistical methods to analyze brain signals.

Publications

1. Atitey, K., Li, J., Papas, B., **Egbon, O. A.**, Li, J. L., Kana, M., Aimola, I., & Anchang, B. Boosting data interpretation with GIBOOST to enhance visualization of complex high-dimensional data. *Briefings in Bioinformatics* **26** (2025) (doi:10.1093/bib/bbaf415).
2. **Egbon**, **O. A.**, Hickey, J. W. & Anchang, B. Fusion of spatiotemporal and network models to prioritize multiscale effects in single-cell perturbations. *Briefings in Bioinformatics* **26** (2025) (doi:10.1093/bib/bbaf277).
3. **Egbon**, **O. A.**, Bogoni, M. A. & Louzada, F. Malaria and anemia in Nigeria: a spatio-temporal modeling and causal analysis. *Statistics in Biosciences,* 1–35 (2025) (doi: 10.1007/s12561-025-09482-9).
4. **Egbon**, **O. A.**, Belachew, A. M., Bogoni, M. A., Gayawan, E. & Louzada, F. Spatial and temporal modeling of conflict related fatality and public health implications in Nigeria. *Scientific Reports* **15,** 1–15 (2025) (doi: 10.1038/s41598-025-98422-0).
5. Omolola, O. F., Gayawan, E. & **Egbon**, **A. O.** Analysis of social and economic factors influencing overweight and obesity among women of childbearing age in Nigeria: A gamlss approach. *Scientific African* **28,** e02673 (2025) (doi: 10.1016/j.sciaf.2025.e02673).
6. **Egbon**, **O. A.** & Gayawan, E. Spatio-temporal modeling of violent conflict and fatality in Nigeria: A point process modeling with SPDE approach. *Statistics, Politics and Policy* **16(1)** (2025) (doi: 10.1515/spp-2024-0028).
7. Ramos, E., **Egbon, O. A.**, Ramos P. L., Rodrigues F. A., & Louzada F. Objective Bayesian analysis for the differential entropy of the Gamma distribution. *Brazilian Journal of Probability and Statistics* **38,** 53–73 (2024).
8. Gayawan, E. Cameron, E., Okitika, T., **Egbon**, **O. A.** & Gething, P. A situational assessment of treatments received for childhood diarrhea in the Federal Republic of Nigeria. *Plos one* **19,** e0303963 (2024) (doi: 10.1371/journal.pone.0303963).
9. Gayawan, E. & **Egbon**, **O. A.** Bayesian multinomial model with inference based on Pólya-Gamma augmentation. *Biostatistics & Epidemiology* **8(1)** (2024) (https://doi.org/10.1080/24709360.2024.2398317).

10. **Egbon**, **O. A.**, Heumann, C., Nascimento, D. C. & Louzada, F. Mixtures of Dirichlet processes for joint spatial modelling of transcranial magnetic stimulation mapping data. *Journal of the Royal Statistical Society Series C: Applied Statistics* **74(1),** qlae042 (2024) (https://doi.org/10.1093/jrsssc/qlae042).
11. **Egbon**, **O. A.**, Belachew, A. M., Bogoni, M. A., Babalola, B. T. & Louzada, F. Bayesian spatio-temporal statistical modeling of violent-related fatality in western and central Africa. *Spatial Statistics* **60,** 100828 (2024) (https://doi.org/10.1016/j.spasta.2024.100828).
12. Ramos, P. L., Jerez-Lillo, N., Segovia, F. A., **Egbon**, **O. A.** & Louzada, F. Power-law distribution in pieces: a semi-parametric approach with change point detection. *Statistics and Computing* **34,** 16 (2024) (https://doi.org/10.1007/s11222-023-10336-x).
13. Gayawan, E. & **Egbon**, **O. A.** Spatio-temporal mapping of stunting and wasting in Nigerian children: A bivariate mixture modeling. *Spatial Statistics* **58,** 100785 (2023) (https://doi.org/10.1016/j.spasta.2023.100785).
14. **Egbon**, **O. A.**, Nascimento, D. & Louzada, F. Prior elicitation for Gaussian spatial process: An application to TMS brain mapping. *Statistics in Medicine* **42(22)** (2023) (https://doi.org/10.1002/sim.9842).
15. Gayawan, E., **Egbon**, **O. A.** & Adegboye, O. Copula based trivariate spatial modelling of childhood illnesses in Western African countries. *Spatial and Spatio-temporal Epidemiology* **46** (2023) (https://doi.org/10.1016/j.sste.2023.100591).
16. **Egbon**, **O. A.** & Gayawan, E. Modelling the spatial patterns of antenatal care utilization in Nigeria with inference based on Pólya-Gamma mixtures. *Journal of Applied Statistics* **51(5)** (2023) (https://doi.org/10.1080/02664763.2022.2164561).
17. **Egbon**, **O. A.** & Bogoni, M. A., Babalola, B. T., & Louzada, F. Under age five children survival times in Nigeria: a Bayesian spatial modeling approach. *BMC Public Health* **22(1),** 1–17 (2022) (https://doi.org/10.1080/02664763.2022.2164561).
18. **Egbon**, **O. A.**, Belachew, A. M. & Bogoni, M. A. Modeling spatial pattern of anemia and malnutrition co-occurrence among under-five children in Ethiopia: A Bayesian geostatistical approach. *Spatial and Spatio-temporal Epidemiology* **43** (2022) (https://doi.org/10.1016/j.sste.2022.100533).
19. Babalola, B. T., Ogunsakin, R. E., **Egbon**, **O. A.** & Adeleke, A. R. A Simulation Based Comparative Study of Some Tests for Checking Homogeneity of Non-Crossing Survival Curves Under High Censoring Rates. *Journal of Applied Probability and Statistics* **17,** 87–99 (2022).
20. **Egbon**, **O. A.**, Belachew, A. M. & Bogoni, M. A. Risk factors of concurrent malnutrition among children in Ethiopia: a bivariate spatial modeling approach. *All Life* **15(1),** 512–536 (2022) (https://doi.org/10.1080/26895293.2022.2067251).
21. Gayawan, E., **Egbon**, **O. A.** & Adebayo, S. B. Spatial modelling of the joint burden of malaria and anaemia co-morbidity in children: A Bayesian geoadditive perspective. *Communications in Statistics: Case Studies, Data Analysis and Applications* **8(2),** 264–281 (2022) (https://doi.org/10.1080/23737484.2022.2031345).
22. Saire, J. E. C., **Egbon**, **O. A.** & Belachew, A. M. *Social Network Analysis to Surveillance the Impact of Covid-19 in the African continent* in *2021 International Conference on Electrical, Computer and Energy Technologies (ICECET)* (2021), 1–6.
23. Louzada, F., Nascimento, D. C. D. & **Egbon**, **O. A.** Spatial Statistical Models: An Overview under the Bayesian Approach. *Axioms* **10(4)** (2021) (https://doi.org/10.3390/axioms10040307).
24. **Egbon**, **O. A.**, Somo-Aina, O. & Gayawan, E. Spatial Weighted Analysis of Malnutrition Among Children in Nigeria: A Bayesian Approach. *Statistics in Biosciences* **13(3),** 495–523 (2021) (https://doi.org/10.1007/s12561-021-09303-9).

Presentations

1. **Egbon, O. A.**, Anchang, Benedict (September 2025). Spatial-ZEDNet: a zero-inflated graphical modeling of exposure-induced differential expression in spatial transcriptomics. *National Institutes of Health Research Festival*, Bethesda, Maryland, USA
2. **Egbon, O. A.**, Anchang, Benedict (August 2025). Joint Mapping of Spatiotemporal Dynamics of Multiple Perturbation Effects in Single-Cell Data. *Joint Statistical Meetings (JSM), American Statistical Association*, Nashville, Tennessee, USA
3. **Egbon, O. A**. (May 2025). Geospatial statistical modeling: an application to public health and research opportunities, *Workshop on spatial epidemiology in Akure, Nigeria*, [*Online*].
4. **Egbon, O. A.**, Anchang, Benedict (September 2024). Mapping the spatial and temporally differentially expressed proteins driven by melanoma immunotherapy, *NIH Research Festival*, Maryland, USA
5. **Egbon, O. A.**, Anchang, Benedict (January 2024). Fusion of Spatiotemporal and Network Models to Quantify the Impact of Glis3 Knockout in beta cell development, *National Institute of Environmental Health Sciences Science Day*, RTP, Durham, NC, USA.
6. **Egbon, O. A**, Nascimento, D. C., Louzada, F. (March 2022). Spatial Power Prior Elicitation: a new approach for Brain TMS, *16th Brazilian Meeting of Bayesian Statistics and VI Latin American Conference on Statistical Computing Proceedings*, Sao Paulo, Brazil.

Professional development & Training

Mentoring training *2025*

- The National Institutes of Health (NIH) Mentoring Certificate Program

PAIR-UP Imaging Workshop *May 2024*

Marine Biological Laboratory (MBL), Boston, USA

- Trained on the use of Zeiss Lattice Lightsheet microscopes for capturing and studying complex biological systems.

Coursera Courses in Data Science *2022*

- Neural network and deep learning (link)
- Structural machine learning project (link)
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization, and Optimization (link)
- Convolutional Neural Networks (link)

9th Workshop on probabilistic and statistical methods for Industry *February 2022*

Institute of Mathematics and Computer Sciences, University of Sao Paulo, Brazil.

- Trained on developing probability measures for complex systems.

6th Workshop of Mathematical Solution for Industrial problems *March 2021*

Center of Mathematical Sciences Applied to Industry, CeMEAI, Sao Carlos, Brazil.

- Solved practical problems for the Brazilian multinational corporation in the petroleum industry (Petrobras).
- Proposed solutions to industrial problems and designed a user interface software for the solution.

Workshop on Distributional Regression Models For Analysis of High Dimensional Spatial Data. *June 2018*

Laboratory for Interdisciplinary Statistical Analysis, Federal University of Technology, Akure, Nigeria.

- Developed Bayesian statistical models to solve practical problems in health sciences in Nigeria.
- Present findings in a formal presentation to governmental organizations to assist in policymaking.

Community Outreach

Volunteer, DNA Day Outreach Program, Wake Young Men's Leadership Academy, Raleigh, NC, USA. *April 2025*

- Led interactive sessions for over **40** high school students on forensic DNA analysis, including hands-on DNA extraction from biological specimens.
- Introduced students to real-world applications of molecular biology through demonstrations and discussions.
- Shared personal research experiences at NIEHS to inspire interest in STEM and biomedical research careers.

Tutor, Statistics, Federal University of Technology, Akure, Akure, Nigeria *July — December 2017*

- Organize free tutorial sessions on Statistics courses for over 40 first- and second-year undergraduate students.
- Initiate interactive sessions that follow lectures and allow for deeper engagement with the course material, questions, and assignments.
- Prepare students for semester exams.

Tutor, Mathematics, Comprehensive High School, Yagba West, Nigeria *September 2014 – April 2015*

- Delivered mathematics instruction to over **100** high school students, fostering foundational skills in algebra, geometry, and quantitative reasoning.
- Adapted teaching methods to accommodate varying skill levels and learning styles, contributing to improved student performance in national examinations.

Computational Skills

Statistical tools:	R, Python, STATA
Database management:	MySQL and POWER BI
Applications:	LaTeX, spreadsheet, and presentation software

Language

English: Fluent in Writing, Speaking, and Listening.
Portuguese: Intermediate in Writing, Speaking, and Listening.